Machine Learning for NLP

NLP II 2026

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Technology

- Automatic text processing
  - Collect the data from various platforms
  - Word segmentation, text cleaning, and normalization
- **Machine Learning**
  - **The intelligent algorithm of NLP**

Machine Learning Techniques in NLP

- Supervised Learning
- Unsupervised Learning
- Transfer Learning

Machine Learning Techniques in NLP

- **Supervised Learning**
- Unsupervised Learning
- Transfer Learning

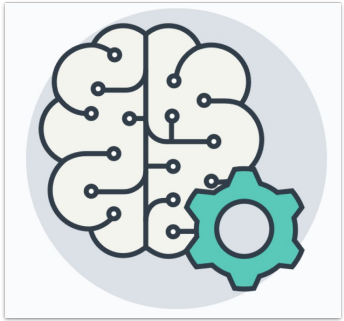
Supervised ML algorithm is a process of learning the relationship between input and output (label).

Supervised ML algorithms learn from examples

Training data

Input	Output

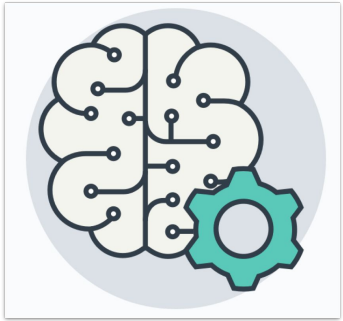
Machine Learning Algorithm



Training data

Input	Output
"สวัสดีครับ"	สวัสดี | ครับ
"ถามหน่อยครับ"	ถาม | หน่อย | ครับ
	...
"ขอโทษค่ะ อะไรนะคะ"	ขอโทษ | ค่ะ | อะไร | นะ | คะ

Machine Learning Algorithm

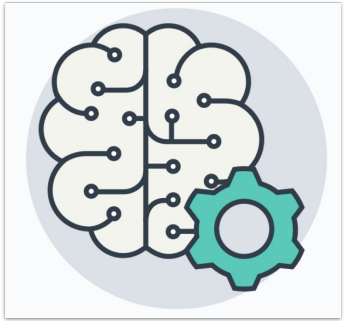


'Training' a word segmenter

Training data

Input	Output
"อาหารเหมาะสำหรับสัตว์เลี้ยงเท่านั้น" "Food here is only suitable for your pets"	😣
"ไกลแค่ไหนก็ต้องมาทาน" "I would drive for hours for the food here"	😀
...	...
"พนักงานบริการเต็มใจมาก" "Their service attitude is overwhelming"	😣

Machine Learning Algorithm



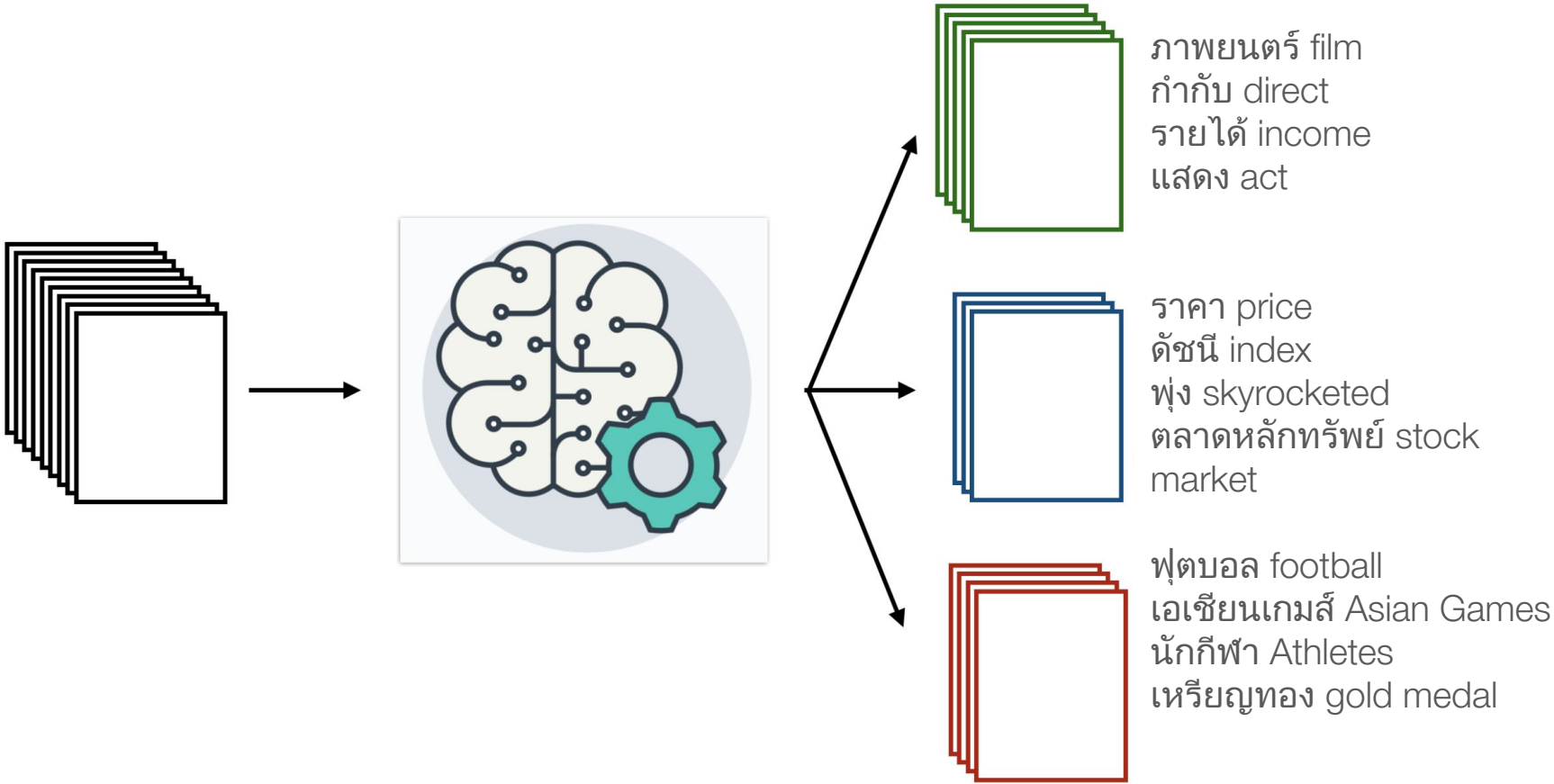
'Training' a sentiment analyzer

To train a model =
To make the machine learn from the data

Machine Learning Techniques in NLP

- Supervised Learning
- **Unsupervised Learning**
- Transfer Learning

Unsupervised ML algorithm is a process of learning the patterns in the input such as natural grouping


ภาพยนตร์ film
กำกับ direct
รายได้ income
แสดง act
ราคา price
ดัชนี index
พุ่ง skyrocketed
ตลาดหลักทรัพย์ stock market
ฟุตบอล football
เอเชียนเกมส์ Asian Games
นักกีฬา Athletes
เหรียญทอง gold medal

Why do prices rise in poor countries?

"Money printing and weak currencies make things more expensive in financially weak nations."

Low-income nations often see rising expenses due to problems with supply and demand.

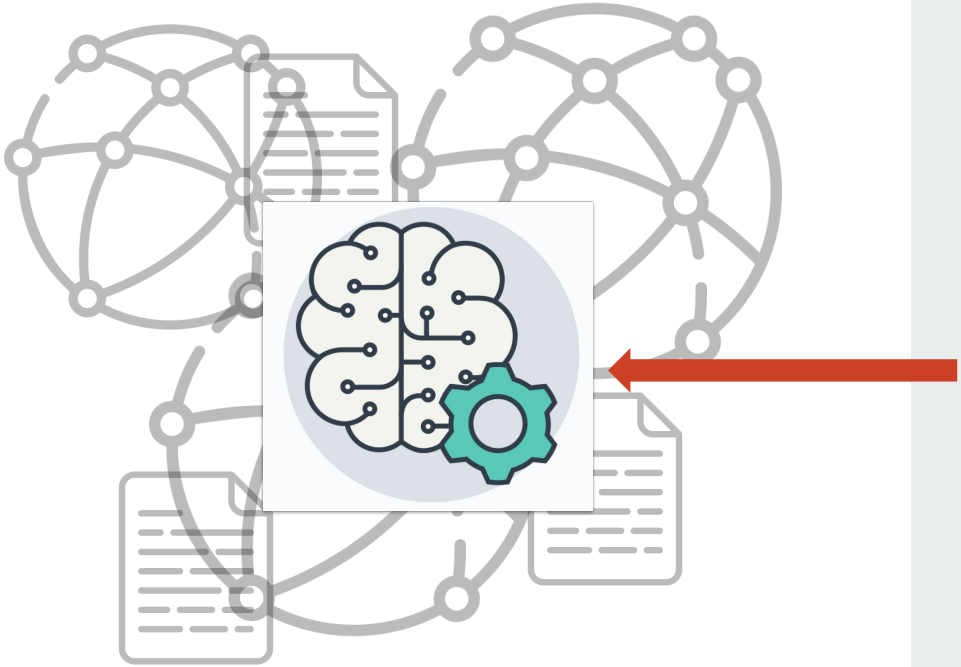
Prices rise in all poor countries

High debt and bad government policies cause price increases in poorer economies.

An unsupervised ML model group the query with the 'right' documents

Machine Learning Techniques in NLP

- Supervised Learning
- Unsupervised Learning
- Transfer Learning



Training data

Input	Output
"อาหารเหมาะสำหรับสัตว์เลี้ยงเท่านั้น" "Food here is only suitable for your pets"	😣
"ไกลแค่ไหนก็ต้องมาทาน" "I would drive for hours for the food here"	😀
...	...
"พนักงานบริการเต็มใจมาก" "Their service attitude is overwhelming"	😣

Learning the language from 50GB+++ of text data from the internet

(Pretraining process)

Learning the task from the training data

(Fine-tuning process)

Text Classification

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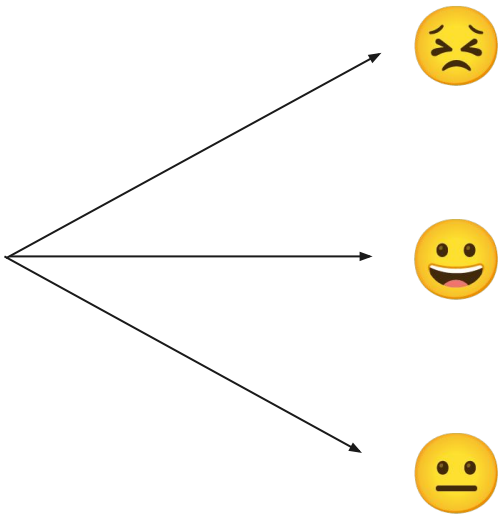
Text Classification

- Text classification is the task of assigning predefined categories or labels (output) to a given piece of text (input), which can be a sentence, a document, or a set of documents.

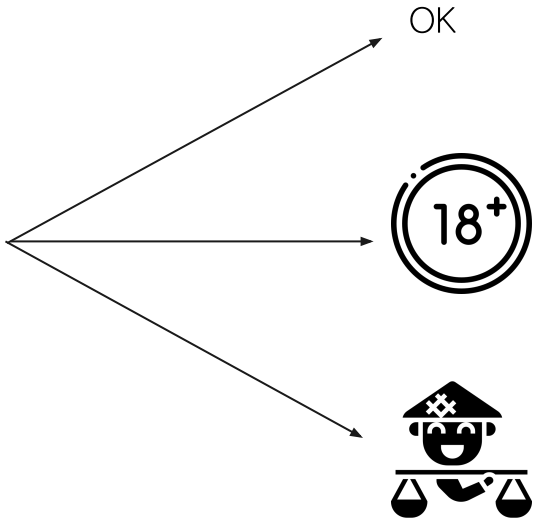
Text input
"อาหารเหมาะสำหรับสัตว์เลี้ยงเท่านั้น" "Food here is only suitable for your pets"
"ไกลแค่ไหนก็ต้องมาทาน" "I would drive for hours for the food here"
...
"พนักงานบริการเต็มใจมาก" "Their service attitude is overwhelming"



Predefined categories or labels

Sentiment Analysis

Text input
"ชื่นชอบผลงานมานานแล้วค่ะ" I've admired your work for a long time
"เงินกู้ด่วน ไม่ต้องค้ำ ดอกเบี้ยต่ำ คลิกเลย" Quick loan! No collateral, low interest—click now!
"ถ้าจะเต้นแค่นี้ กลับบ้านเหอะ เซ็ง" If you're gonna dance like that, just go home. Boring.
...
"เพลงน่าจะชัดกว่านี้ค่ะ แต่เต้นเป๊ะมาก" The music could've been clearer, but the dancing was spot-on.



Predefined categories or labels

Spam classification

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word boundary

non-word boundary

Predefined categories or labels

บุญคุณ|ต้อง|ทดแทน

Word segmentation result

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word boundary

non-word boundary

Predefined categories or labels

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word boundary

non-word boundary

Predefined categories or labels

Word segmentation

Other examples

Automatic Essay Grading

- Input: essay text
- Output: {A, B, C, D}

Language Identification

- Input: text
- Output: {FR, EN, TH, ZH, …}

Fake news detection

- Input: news text
- Output: fake or not-fake

Intent classification

- Input: utterance
- Output: {apology, request, complaint, … }

Types of Classifiers

- Rule-based classifier
- Zero-shot LLM classifier
- ML-based classifier

Rule-based classifier

- A classifier that relies on a set of predefined rules to assign labels to text such as keywords, regular expressions, and lexicons.
- A lexicon (or dictionary or word list) is a collection of words and/or phrases optionally along with their meanings or attributes.

Spam Classifier - keyword-based classifier

```
def spam_classify(text):
    keywords = {'viagra', 'lottery', 'free money'}
    for word in text:
        if word in key_words:
            return 'SPAM'
    return 'NOT SPAM'
```

Email classifier -- regex-based

```
def email_classify(token):
    pattern = re.compile('^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$')
    return pattern.match(token)
```

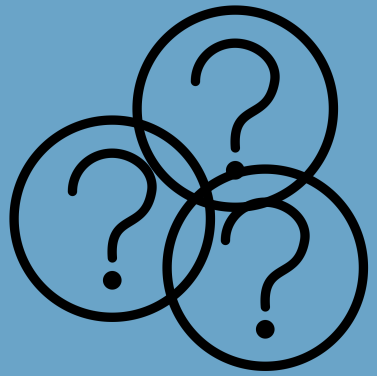
Name recognition - Lexicon-based

```
def recognize_name(token):
    # Where do we get the list of names from?
    name_list = read_lexicon_from_file('my_name_lexicon.txt')
    if token in name_list:
        return 'NAME'
    return 'NOT NAME'
```

Discuss with your neighbor

- Write a rule-based classifier for detecting English vs French vs Spanish
- Write a rule-based classifier for recognizing typically Thai male name vs typically Thai female name

What are the pros and cons of rule-based classifiers?



Pros and cons of rule-based classifier

Pros

- Simplicity
- Transparency
- Scalability

Cons

- Limited performance
- Sensitive to noise
- Labor intensive

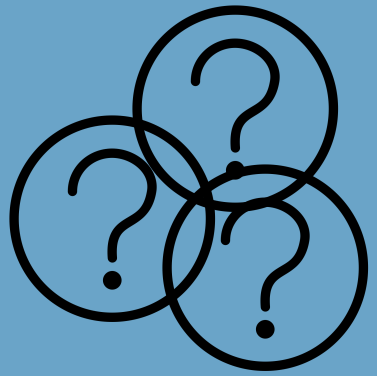
Types of Classifiers

- Rule-based classifier
- **Zero-shot LLM classifier**
- ML-based classifier

Zero-shot Large Language Model Classifier

- Large language models take advantage of transfer learning and emergent properties (good things that come up unexpectedly) from the training process.
- Zero-shot means the system performs the task without
  - direct instruction: we code up the program directly
  - direct labeled supervision: we provide samples for learning the task

Why do people really want zero-shot classification?



LLM can perform typical (zero-shot) NLP tasks

- Text classification e.g. sentiment analysis, intent classification
- Linguistic analysis e.g. word segmentation, part-of-speech tagging
- Extract named entities
- Text translation
- Text summarization
- Reading comprehension
- Grammatical error correction
- Paraphrasing and improving text quality

LLM classifier - Example I

What is the sentiment of the following product review, which is delimited with triple backticks?

Give your answer as a single word, either "positive" or "negative".

Review text: ```Needed a nice lamp for my bedroom, and this one had additional storage and not too high of a price point. Got it fast. The string to our lamp broke during the transit and the company happily sent over a new one. Came within a few days as well. It was easy to put together. I had a missing part, so I contacted their support and they very quickly got me the missing piece! Lumina seems to me to be a great company that cares about their customers and products!!```

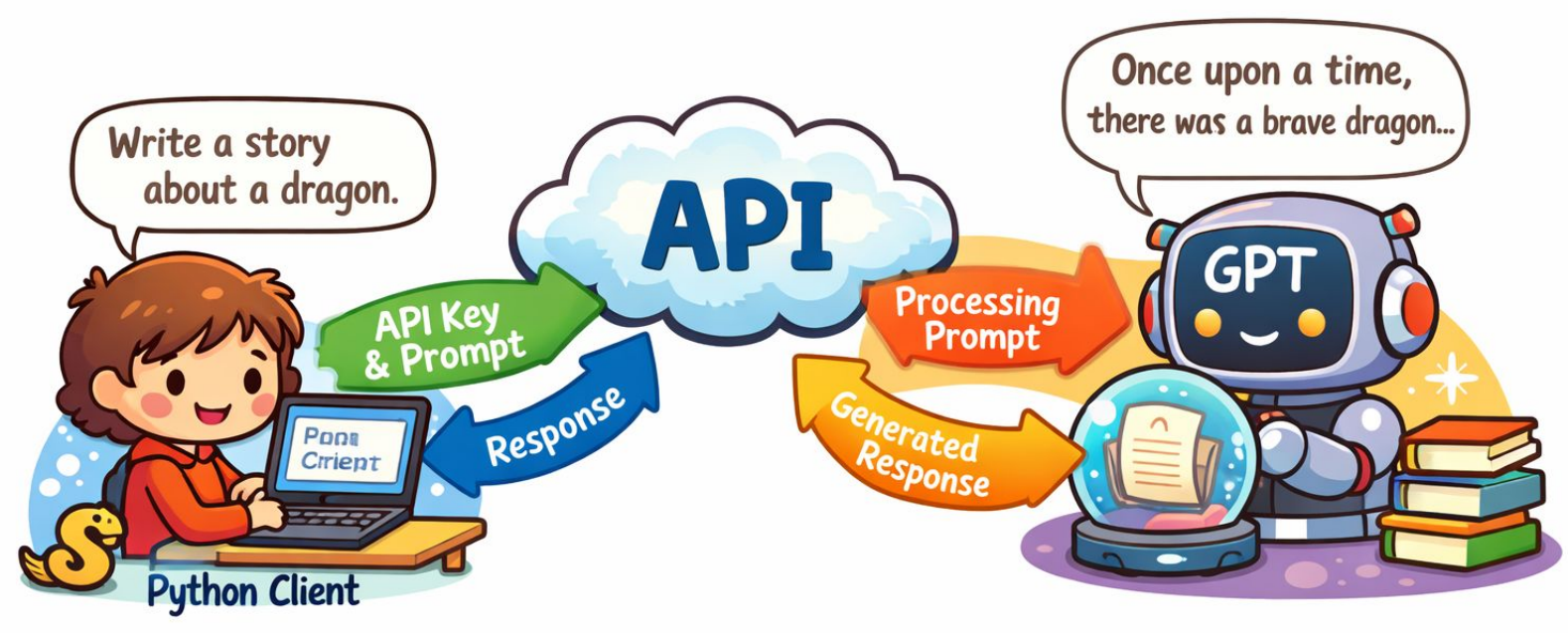
LLM classifier - Example II

Is the writer of the following review expressing anger? The review is delimited with triple backticks. Give your answer as either yes or no.

Review text: ```Needed a nice lamp for my bedroom, and this one had additional storage and not too high of a price point. Got it fast. The string to our lamp broke during the transit and the company happily sent over a new one. Came within a few days as well. It was easy to put together. I had a missing part, so I contacted their support and they very quickly got me the missing piece! Lumina seems to me to be a great company that cares about their customers and products!!```

LLM classifier - Example III

```
We want to classify the message marked by triple backticks into one of the ten labels explained below:
Greeting: Recognizing when a user is saying hello or initiating a conversation.
Goodbye: Identifying when a user is ending the conversation or saying goodbye.
FAQ: Handling frequently asked questions about a product, service, or topic.
Order Placement: Recognizing when a user wants to place an order for a product or service.
Booking: Handling requests for reservations or appointments, such as booking a hotel room or scheduling a meeting.
Information Request: Recognizing when a user is seeking specific information, such as contact details, opening hours,
or pricing.
Feedback: Identifying when a user wants to provide feedback or report an issue.
Small Talk: Engaging in casual conversation with the user, like discussing the weather or asking how they are doing.
Complaints: Recognizing when a user is expressing dissatisfaction or lodging a complaint.
User Assistance: Handling requests for assistance or troubleshooting, such as helping with technical issues or guiding
the user through a process.
Give the output in json format the key is 'label', and the other key is 'reason' to give reason for the classification
```The staff at the lobby is very lazy. They never answer the phone.```
```
```



- You send your prompt over the internet to OpenAI using the client, which helps you securely access the API. OpenAI API requires an API key to access the service.
- OpenAI GPT models return the response through the internet back to your computer

```python
def analyze_sentiment(text):
    sentiment_analysis_prompt = f"""
    Analyze the sentiment of the text delimited by <text> and </text> tags.
    The sentiment must be one of the following: positive, negative, or neutral.

    Output must be label string only

    <text>
    {text}
    </text>
    """
    messages = [
            {"role": "user", "content": sentiment_analysis_prompt}
        ]
    response = openai.chat.completions.create(
        model="gpt-4o-mini",
        messages=messages,
    )
    return response.choices[0].message.content

sentiment = analyze_sentiment("บริษัทจะต้องเน้นการสร้างความเชื่อมั่นให้กับลูกค้าให้มากขึ้น")
```

```python
def analyze_sentiment_with_reason(text):
    sentiment_analysis_prompt = f"""
    Analyze the sentiment of the text delimited by <text> and </text> tags.
    The sentiment must be one of the following: positive, negative, or neutral.

    Output must be json with keys:
        reason: the reason for the sentiment
        sentiment: the sentiment label

    <text>
    {text}
    </text>
    """
    messages = [
            {"role": "user", "content": sentiment_analysis_prompt}
        ]
    response = openai.chat.completions.create(
        model="gpt-4o-mini",
        messages=messages,
        response_format={ "type": "json_object" }
    )
    json_response = json.loads(response.choices[0].message.content)
    return json_response

sentiment = analyze_sentiment("บริษัทจะต้องเน้นการสร้างความเชื่อมั่นให้กับลูกค้าให้มากขึ้น")
```

Zero-shot large language model classifier

- LLMs are good at a broad range of tasks but do not excel at any of them.
- LLMs are very useful for tasks that samples are difficult to obtain.
- LLMs are expensive (why?)

Types of Classifiers

- Rule-based classifier
- Zero-shot LLM classifier
- **ML-based classifier**

ML-based classifier

- An ML-based classifier rely on supervised learning methods that learn from labeled text data to classify new unseen text data.
- Popular ML models
  - Logistic Regression
  - Deep learning models

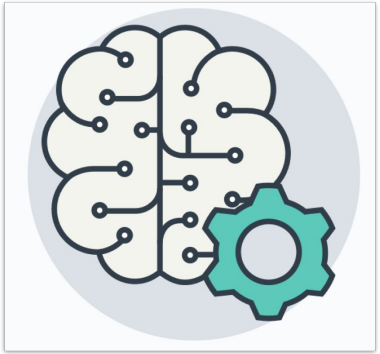
4 Steps for Supervised Learning

Input	Output
"อาหารเหมาะสำหรับสัตว์เลี้ยงเท่านั้น"	😣
"ไกลแค่ไหนก็ต้องมาทาน"	😀
...	...
"พนักงานบริการเต็มใจมาก"	😣

Data preparation

f1	f2	f3	label
0	1	3	บวก
-1.0	0	4	ลบ
1	0	3	กลาง
1	1	4	บวก
...	...	...	...

Feature engineering



Model training

Evaluation

1. Data Preparation

Text input
"อาหารเหมาะสำหรับสัตว์เลี้ยงเท่านั้น" "Food here is only suitable for your pets"
"ไกลแค่ไหนก็ต้องมาทาน" "I would drive for hours for the food here"
...
"พนักงานบริการเต็มใจมาก" "Their service attitude is overwhelming"

Choosing data sources

Defining instances

Defining labels

Data Annotation

Data needs to be labeled (annotated) by humans. This process is often costly and slow.

Input	Output
"อาหารเหมาะสำหรับสัตว์เลี้ยงเท่านั้น" "Food here is only suitable for your pets"	😣
"ไกลแค่ไหนก็ต้องมาทาน" "I would drive for hours for the food here"	😀
...	...
"พนักงานบริการเต็มใจมาก" "Their service attitude is overwhelming"	😣

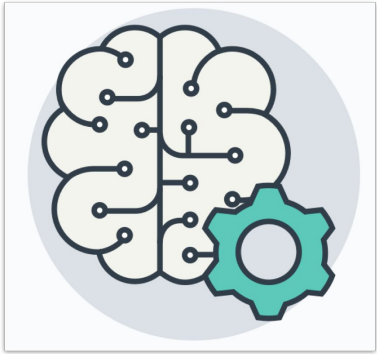
4 Steps for Supervised Learning

Input	Output
"อาหารเหมาะสำหรับสัตว์เลี้ยงเท่านั้น"	😣
"ไกลแค่ไหนก็ต้องมาทาน"	😀
...	...
"พนักงานบริการเต็มใจมาก"	😣

Data preparation

f1	f2	f3	label
0	1	3	บวก
-1.0	0	4	ลบ
1	0	3	กลาง
1	1	4	บวก
...	...	...	...

Feature engineering



Model training

Evaluation

Feature Engineering

Extracting relevant features (numbers or signal) from the text data that are indicative of the label.

Each text instance is converted into a feature vector.

f1	f2	f3	label
0	1	3	บวก
-1.0	0	4	ลบ
1	0	3	กลาง
1	1	4	บวก
...	...	...	...

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!

it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
…	…

Bag-of-word features

Taken from Martin and Jurafsky 3rd ed.

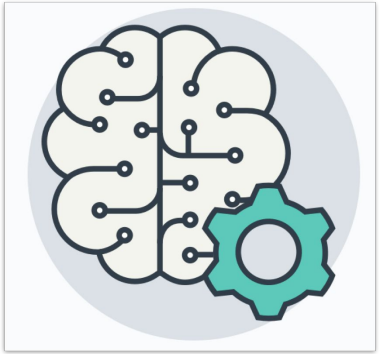
4 Steps for Supervised Learning

Input	Output
"อาหารเหมาะสำหรับสัตว์เลี้ยงเท่านั้น"	😣
"ไกลแค่ไหนก็ต้องมาทาน"	😀
...	...
"พนักงานบริการเต็มใจมาก"	😣

Data preparation

f1	f2	f3	label
0	1	3	บวก
-1.0	0	4	ลบ
1	0	3	กลาง
1	1	4	บวก
...	...	...	...

Feature engineering

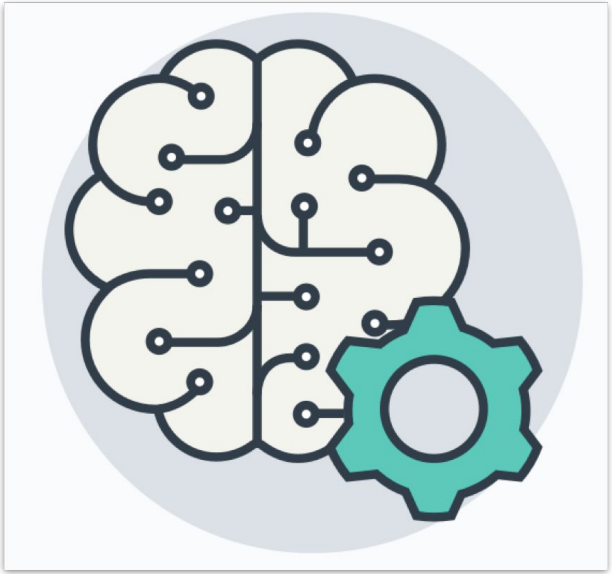


Model training

Evaluation

Model Training

Feeding the training data to the model for the model to learn the relationship between features and labels.



Train-validation-test data split

The data must first be split into three subsets:

- Training set -- for training models
- Validation set (holdout set or development set) -- for iteratively evaluating models
- Test set -- for final evaluation of the model

For example, 60:20:20 Split

Training set

Validation set

Test set

idx	Date	Text	Label
1	31 Dec 2019	xxx	yy
2	1 Jan 2020	xxx	yy
3	4 Jan 2020	xxx	yy
4	7 Jan 2020	xxx	yy
5	8 Jan 2020	xxx	yy
6	8 Jan 2020	xxx	yy
7	9 Jan 2020	xxx	yy
8	9 Jan 2020	xxx	yy
9	9 Jan 2020	xxx	yy
10	11 Jan 2020	xxx	yy

Bias in data preparation

- If the data is ordered (e.g., by time or some natural sequence), then splitting without shuffling might result in the training set containing only older data and the test set containing newer data.

For example, 60:20:20 Split

Training set

Validation set

Test set

idx	Date	Text	Label
1	31 Dec 2019	xxx	yy
2	1 Jan 2020	xxx	yy
3	4 Jan 2020	xxx	yy
4	7 Jan 2020	xxx	yy
5	8 Jan 2020	xxx	yy
6	8 Jan 2020	xxx	yy
7	9 Jan 2020	xxx	yy
8	9 Jan 2020	xxx	yy
9	9 Jan 2020	xxx	yy
10	11 Jan 2020	xxx	yy

Shuffled

Training set

Validation set

Test set

idx	Date	Text	Label
4	7 Jan 2020	xxx	yy
7	9 Jan 2020	xxx	yy
1	31 Dec 2019	xxx	yy
10	11 Jan 2020	xxx	yy
5	8 Jan 2020	xxx	yy
2	1 Jan 2020	xxx	yy
6	8 Jan 2020	xxx	yy
9	9 Jan 2020	xxx	yy
8	9 Jan 2020	xxx	yy
3	4 Jan 2020	xxx	yy

Deduplication

- Deduplication is removal of identical or nearly identical examples that appear multiple times within the same dataset or across different splits
- Why?
  - Duplicates can give the false impression of having more diverse data than is actually present.
  - Duplicates in the dev/test sets can bias performance metrics because the model is essentially being tested on the same input multiple times.

Model training

- We train the model on the training set. Each model has its own formula for training the model parameters.
- We evaluate the model on the dev set. Each model has its own way of using the trained parameters. This process is called 'inference' (the model infers the labels from the text)

4 Steps for Supervised Learning

Input	Output
"อาหารเหมาะสำหรับสัตว์เลี้ยงเท่านั้น"	😣
"ไกลแค่ไหนก็ต้องมาทาน"	😀
...	...
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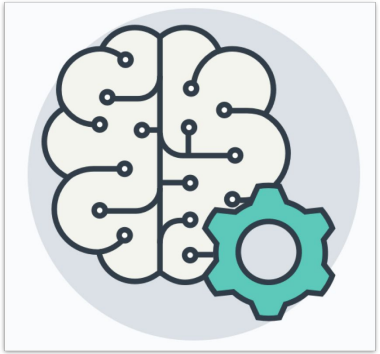
Data preparation

f1	f2	f3	label
0	1	3	บวก
-1.0	0	4	ลบ
1	0	3	กลาง
1	1	4	บวก
...	...	...	...

Feature engineering

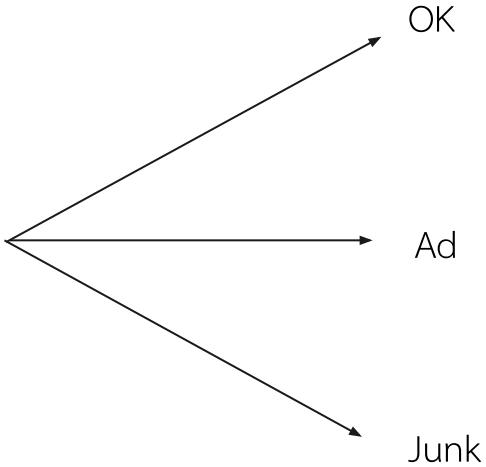


Model training

Evaluation

Example - Spam classification

Text input
"ชื่นชอบผลงานมานานแล้วค่ะ" I've admired your work for a long time
"เงินกู้ด่วน ไม่ต้องค้ำ ดอกเบี้ยต่ำ คลิกเลย" Quick loan! No collateral, low interest—click now!
"ถ้าจะเต้นแค่นี้ กลับบ้านเหอะ เซ็ง" If you're gonna dance like that, just go home. Boring.
...
"เพลงน่าจะชัดกว่านี้ค่ะ แต่เต้นเป๊ะมาก" The music could've been clearer, but the dancing was spot-on.



Prediction on dev set

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	...	97	98	99	100
Gold standard คำตอบจริง	J	J	J	J	Ad	Ad	Ad	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK

Accuracy

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	...	97	98	99	100
Gold standard คำตอบจริง	J	J	J	J	Ad	Ad	Ad	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK

Accuracy = 93 / 100

Now compute the accuracy of System A and B

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	...	97	98	99	100
Gold standard คำตอบจริง	J	J	J	J	Ad	Ad	Ad	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	J	J	J	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	OK	OK	...	OK	OK	OK	OK

Accuracy of System A = 93 / 100
Accuracy of System B = 93 / 100

Precision and Recall

- Precision of label A: If classifier predicts 'A', can we trust it's actually A?

  # A correctly predicted / # predicted A

- Recall of label A: How many true 'A' are actually detected?

  # A correctly predicted / # true A

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	...	97	98	99	100
Gold standard คำตอบจริง	J	J	J	J	Ad	Ad	Ad	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	J	J	J	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	OK	OK	...	OK	OK	OK	OK

	Precision	Recall
OK email	93/100	93/93
Junk Mail (J)	NA	0/4
Advertisement (Ad)	NA	0/3

F1 score

- Precision and recall are calculated for each label. F1 score is the geometric mean of Precision (P) and Recall (R)

$$\text{F1 of class A} = 2PR / (P+R)$$

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	...	97	98	99	100
Gold standard คำตอบจริง	J	J	J	J	Ad	Ad	Ad	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	J	J	J	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	OK	OK	...	OK	OK	OK	OK

	Precision	Recall	F1
OK email	0.93	1.0	0.96
Junk Mail (J)	NA	0/4	0
Advertisement (Ad)	NA	0/3	0

F1 of class OK = 2PR / (P+R)

= 2 * (0.93)* (1.0) / (0.93 + 1.0)

nlp@chula

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	...	97	98	99	100
Gold standard คำตอบจริง	J	J	J	J	Ad	Ad	Ad	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	J	J	J	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	OK	OK	...	OK	OK	OK	OK

		Precision		Recall	F1
OK email	86/86	1.0	86/93	0.9247	0.9609
Junk Mail (J)	3/3	1.0	3/4	0.75	0.8571
Advertisement (Ad)	3/10	0.3	3/3	1	0.4615

nlp@chula

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	...	97	98	99	100
Gold standard คำตอบจริง	J	J	J	J	Ad	Ad	Ad	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	J	J	J	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	OK	OK	...	OK	OK	OK	OK

	Precision	Recall	F1
OK email	0.93	1.0	0.96
Junk Mail (J)	NA	0/4	0
Advertisement (Ad)	NA	0/3	0

System A

	Precision	Recall	F1
OK email	1.0	0.9247	0.9609
Junk Mail (J)	1.0	0.75	0.8571
Advertisement (Ad)	0.3	1	0.4615

System B

Precision and Recall

- Precision of label A: If classifier predicts 'A', can we trust it's actually A?
  - We want high precision when we want to really trust the system.
- Recall of label A: How many 'A' are actually detected?
  - We want high recall when we do not want to miss that label.

Consider this scenario

You built a classifier that classifies whether a TCAS portfolio deserves admission

Suppose the cost for admitting an 'unqualified' candidate is $100
Suppose the cost for missing a qualified candidate is $10000

System A:

- Precision is 0.3 → Only 30% of the admitted students are actually qualified.
- Recall is 0.8 → Only 80% of the 'qualified' students are admitted.

System B:

- Precision is 0.9 → Only 90% of the admitted students are actually qualified.
- Recall is 0.2 → Only 20% of the 'qualified' students are admitted.

Consider this scenario

You built a classifier that classifies whether a review is fake and should be deleted

Suppose the cost for wrongly deleting a fake review is $1000
Suppose the cost for putting the fake review online is $10

System A:

- Precision is 0.3 → Only 30% of the deleted reviews are actually fake.
- Recall is 0.8 → Only 80% of the fake reviews are deleted.

System B:

- Precision is 0.9 → Only 90% of the deleted reviews are actually fake.
- Recall is 0.2 → Only 20% of the fake reviews are deleted.

Macro-average precision, recall, and F1

	Precision	Recall	F1
OK email	1.0	0.9247	0.9609
Junk Mail (J)	1.0	0.75	0.8571
Advertisement (Ad)	0.3	1	0.4615
Macro-average	0.76	0.89	0.75

Confusion Matrix

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	...	97	98	99	100
Gold standard คำตอบจริง	J	J	J	J	Ad	Ad	Ad	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	...	OK	OK	OK	OK
Prediction เครื่องเดามา	J	J	J	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	OK	OK	...	OK	OK	OK	OK

Actual (ความเป็นจริง)

Predicted (เครื่องบอกมา)	Junk	Ad	OK
Junk	3	0	0
Ad	1	3	6
OK	0	0	87

Actual (ความเป็นจริง)

Predicted (เครื่องบอกมา)	Junk	Ad	OK
Junk	3	0	0
Ad	1	3	6
OK	0	0	87

	Junk	Ad	OK
Recall	3 / (3 + 1 + 0)		

Actual (ความเป็นจริง)

Predicted (เครื่องบอกมา)	Junk	Ad	OK
Junk	3	0	0
Ad	1	3	6
OK	0	0	87

	Junk	Ad	OK
Recall	3 / (3 + 1 + 0)	3 / (3 + 0 + 0)	87 / (0 + 6 + 87)

Actual (ความเป็นจริง)

Predicted (เครื่องบอกมา)	Junk	Ad	OK
Junk	3	0	0
Ad	1	3	6
OK	0	0	87

	Junk	Ad	OK
Recall	0.75	1.0	0.93

Actual (ความเป็นจริง)

Predicted (เครื่องบอกมา)	Junk	Ad	OK
Junk	3	0	0
Ad	1	3	6
OK	0	0	87

	Precision
Junk	3 / (3 + 0 + 0)
Ad	
OK	

	Junk	Ad	OK
Recall	0.75	1.0	0.93

Actual (ความเป็นจริง)

Predicted (เครื่องบอกมา)	Junk	Ad	OK
Junk	3	0	0
Ad	1	3	6
OK	0	0	87

	Precision
Junk	3 / (3 + 0 + 0)
Ad	3 / (1 + 3 + 6)
OK	

	Junk	Ad	OK
Recall	0.75	1.0	0.93

Actual (ความเป็นจริง)

Predicted (เครื่องบอกมา)	Junk	Ad	OK
Junk	3	0	0
Ad	1	3	6
OK	0	0	87

	Precision
Junk	3 / (3 + 0 + 0)
Ad	3 / (1 + 3 + 6)
OK	87 / (0 + 0 + 87)

	Junk	Ad	OK
Recall	0.75	1.0	0.93

Actual (ความเป็นจริง)

Predicted (เครื่องบอกมา)	Junk	Ad	OK
Junk	3	0	0
Ad	1	3	6
OK	0	0	87

	Precision
Junk	1
Ad	0.3
OK	1

	Junk	Ad	OK
Recall	0.75	1.0	0.93

Actual (ความเป็นจริง)

Predicted (เครื่องบอกมา)	Junk	Ad	OK
Junk	3	0	0
Ad	1	3	6
OK	0	0	87

	Precision
Junk	1
Ad	0.3
OK	1

	Junk	Ad	OK
Recall	0.75	1.0	0.93

Accuracy (3 + 3 + 87) / (3 + 3 + 87 + 1 + 6)

Actual (ความเป็นจริง)

Predicted (เครื่องบอกมา)	Junk	Ad	OK
Junk	3	0	0
Ad	1	3	6
OK	0	0	87

	Precision
Junk	1
Ad	0.3
OK	1

	Junk	Ad	OK
Recall	0.75	1.0	0.93

Accuracy	0.93

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	…	97	98	99	100
Gold standard คำตอบจริง	J	J	J	J	Ad	Ad	Ad	OK	OK	OK	OK	OK	OK	OK	OK	…	OK	OK	OK	OK
Prediction เครื่องเดามา	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	…	OK	OK	OK	OK
Prediction เครื่องเดามา	J	J	J	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	OK	OK	…	OK	OK	OK	OK

Actual (ความเป็นจริง)

Predicted (เครื่องบอกมา)	Junk	Ad	OK
Junk	3	0	0
Ad	1	3	6
OK	0	0	87

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	…	97	98	99	100
Gold standard คำตอบจริง	J	J	J	J	Ad	Ad	Ad	OK	OK	OK	OK	OK	OK	OK	OK	…	OK	OK	OK	OK
Prediction เครื่องเดามา	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	…	OK	OK	OK	OK
Prediction เครื่องเดามา	J	J	J	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	Ad	OK	OK	…	OK	OK	OK	OK

		Actual (ความเป็นจริง)		
		Junk	Ad	OK
Predicted (เครื่องบอกมา)	Junk	0	0	0
	Ad	0	0	0
	OK	4	3	93

Confusion Matrix

Accuracy, precision, recall, and F1 can be computed from a confusion matrix

Useful for analyzing errors

		Actual (ความเป็นจริง)		
		Junk	Ad	OK
Predicted (เครื่องบอกมา)	Junk	3	0	0
	Ad	1	3	6
	OK	0	0	87

Conclusion

- We learn how to use rule-based models and supervised learning models for text classification, which can solve many NLP tasks.
- Supervised learning tasks involve task formulation, feature engineering, model training, and evaluation.
- Machine learning models differ in how they learn and infer.
- Evaluation is very important in understanding the performance of various models